

 OPENAPI 3.0 SPECIFICATION

TicketWave — Functional Specification

An overview of all 69 annotated REST endpoints across 20 controllers: role-based security, contract definitions, error standards, and multi-tenant scoping.

SECTION 1

API Principles & Exception Standards

Consistent error contracts, precise monetary types, and self-documenting OpenAPI endpoints.

Core API Architecture Standards



Unified Error Response Shape

Enforced across all endpoints via `GlobalExceptionHandler`:

```
{ "status": 404, "error": "BOOKING_NOT_FOUND", "message":  
"...", "timestamp": "..." }
```



Financial & Contract Precision

Monetary Precision: All currency and financial calculation fields strictly mapped as `BigDecimal`.

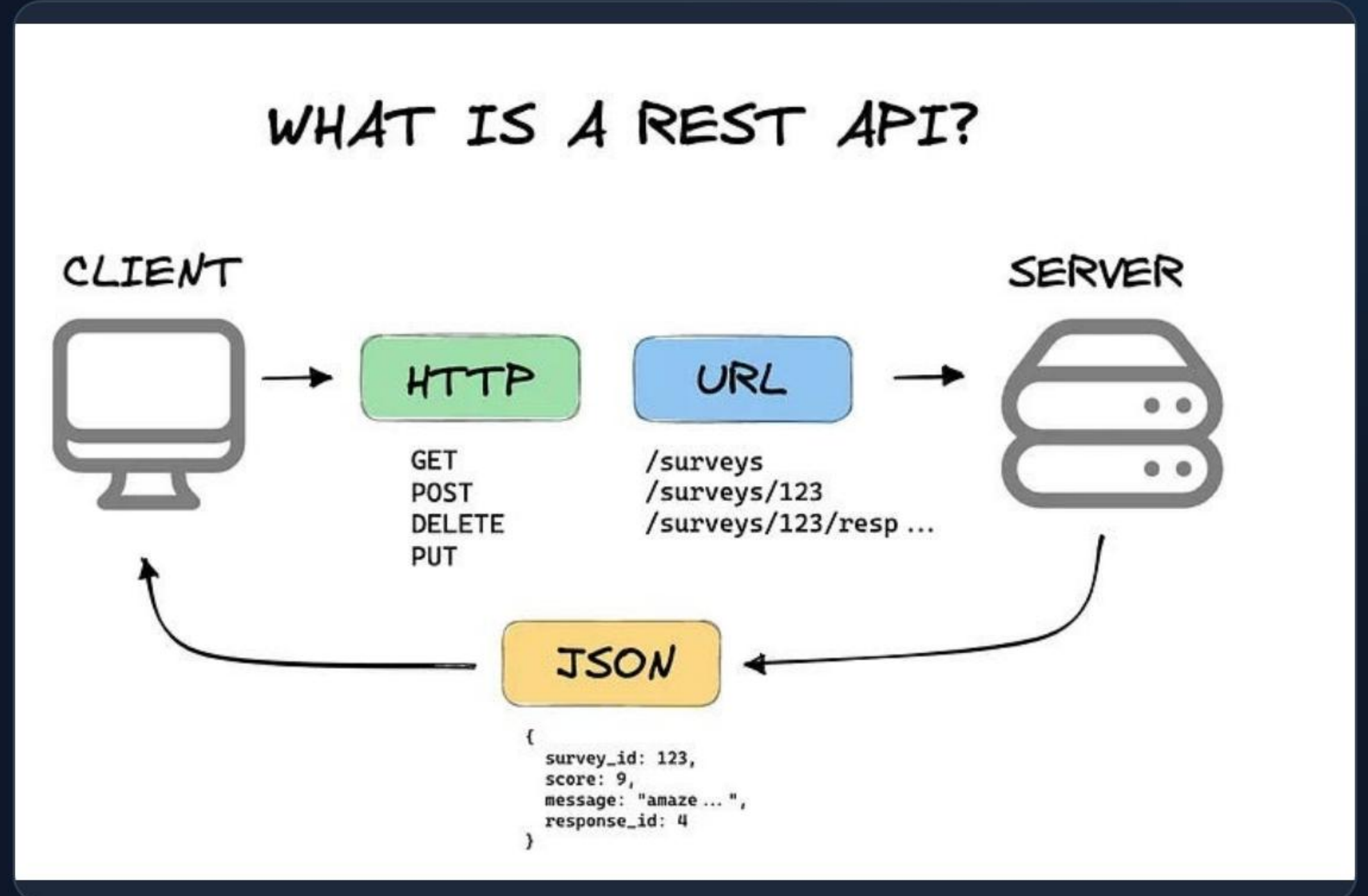
Live OpenAPI Schema: Automatically generated at runtime via SpringDoc at `/v3/api-docs` and `/swagger-ui.html`.

Role-Based Access Control Matrix

Role Level	Token Type	Access Scope & Boundaries
Public	Unauthenticated	Login, registration, schedule search, guest PNR lookup, promo preview
Customer	JWT Bearer	Account-owned seat holds, bookings, payments, refunds, saved passengers
Operator	JWT Bearer (OPERATOR)	Scoped to owned routes, schedules, seat grids, fleet & occupancy reports
Support / Admin	JWT Bearer (SUPPORT)	Global PNR search, refund reviews, fee waiver overrides
Admin	JWT Bearer (ADMIN)	User RBAC, partner credentials, webhooks, audit search, ledger reconciliation
Partner API	OAuth2 Bearer	Short-lived B2B integration token issued via /api/oauth/token

Authentication & Schedule Discovery

- ➔ **Auth Endpoints:** `POST /api/register` (creates CUSTOMER) and `POST /api/login` (returns JWT bearer token).
- 🔍 **Catalog Search:** `GET /api/search` supports optional type, origin, destination, venue, date, and price filters with real-time seat counts.
- 📅 **Seat Maps:** `GET /api/schedules/{id}/seats` returns seat statuses and per-seat `estimatedFare`.



| Customer Booking Lifecycle



1. Seat Holds

`POST /hold` locks seats for a TTL. `DELETE /hold` is a silent, idempotent no-op for non-holders.



2. Booking & Payment

`POST /bookings` creates `INITIATED` reservation. `POST /payments` is **idempotent on reference**.



3. Reschedule & Refund

`reschedule-quote` and `refund-quote` allow risk-free previews before committing changes.

| Guest Servicing & Customer Account



2-Factor Guest PNR Lookup

`GET /api/bookings/pnr/{pnr}/lookup?email=`

Requires both PNR and account email as a second factor, preventing unauthorized itinerary harvesting via brute-force PNR guessing.



Account & Passenger Profiles

`/api/users/me` supports profile management; password changes explicitly demand current password verification.

`/api/passengers` exposes full CRUD operations for saved traveler profiles owned by the caller.

Operator Console & Security Isolation

Defensive Resource Isolation

Operator endpoints check ownership strictly. Any attempt to query or mutate a foreign-owned route or vehicle returns ****404 Not Found**** (never 403) to eliminate resource scanning.

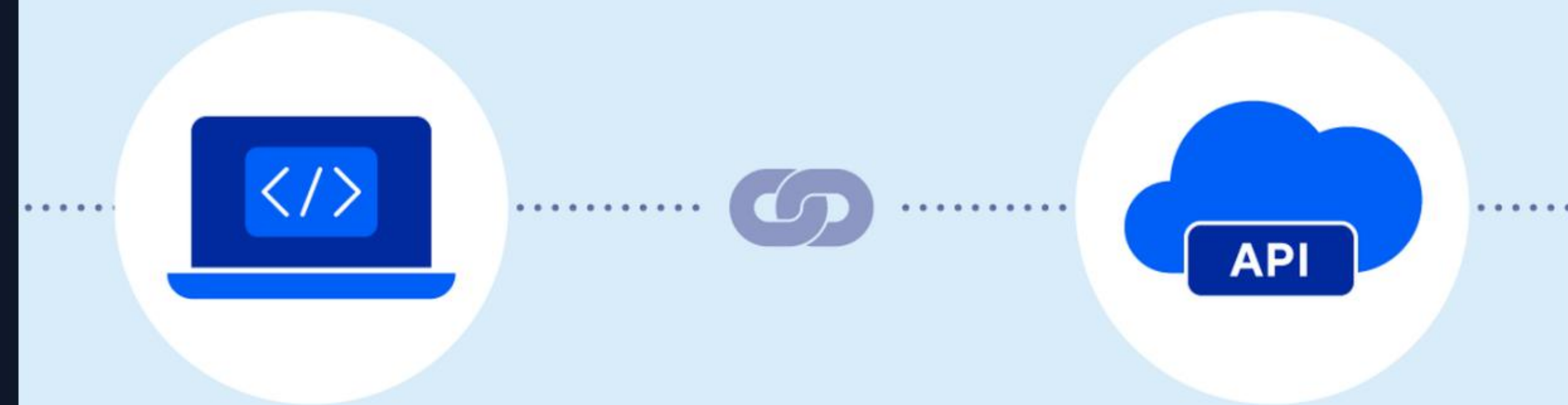
Inventory & Pricing Controls

Full route, schedule, seat grid, vehicle, and driver management. Features `POST /api/fare-rules/bulk` for all-or-nothing CSV fare rule loading.

Revenue & Occupancy Analytics

`GET /api/operator/reports` delivers real-time confirmed booking totals, revenue, and seat occupancy percentages.

work?



Developers

Developers will design your app in a way that it can access data stored in assets/databases via APIs.

API

API will connect with assets/databases to access required info, sending it to the app.

| Admin Console & B2B Integrations



RBAC Provisioning

`POST /api/users` provisions staff roles (`OPERATOR`, `SUPPORT`, `ADMIN`). Reassigning roles protects the last remaining Admin.



Partner OAuth2 & Webhooks

Issues OAuth2 credentials (secret displayed once). Registers outbound webhooks with HMAC-SHA256 signing keys stored in plaintext.



Audit & Reconciliation

`GET /api/audit` searches audit logs; `GET /api/ledger/reconciliation` aggregates payments/refunds over date ranges.

Controller Scope Breakdown

Domain Controller Category	Controllers	Endpoints	Key Operations Handled
Authentication & User Account	4 Controllers	11 Endpoints	Register, login, profile, password change, preferences, passengers
Catalog, Search & Schedules	2 Controllers	5 Endpoints	Public schedule search, seat maps, hold management
Bookings, Payments & Refunds	2 Controllers	15 Endpoints	Booking creation, payment 3DS, reschedule/refund quotes, processing
Operator Console	6 Controllers	16 Endpoints	Routes, schedules, seats, vehicles, drivers, bulk fare rules, reports
Admin Console & Partner API	6 Controllers	22 Endpoints	User RBAC, promo validation, partner OAuth2, webhooks, audit, ledger

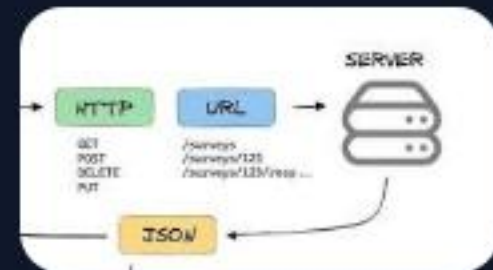
✔ TICKETWAVE OPENAPI COMPLETE

Questions & Discussion

69 Endpoints Annotated | Live Spec: </v3/api-docs> & </swagger-ui.html>

`com.ticketwave.*.controller` | 20 RestControllers Annotated

Image Sources



https://miro.medium.com/1*m3jEkdc9SKTK6vNPhRHCqg.jpeg

Source: medium.com



[https://a.storyblok.com/f/47007/2400x1314/cb325d2266/api-glossary.png/m/2880x0/filters:quality\(80\)](https://a.storyblok.com/f/47007/2400x1314/cb325d2266/api-glossary.png/m/2880x0/filters:quality(80))

Source: www.adjust.com